UNFOLD: 3-D U-Net, 3-D CNN, and 3-D Transformer-Based Hyperspectral Image Denoising

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***Abstract*— Hyperspectral images (HSIs) encompass data across numerous spectral bands, making them valuable in various practical fields such as remote sensing, agriculture, and marine monitoring. Unfortunately, inevitable noise introduction during sensing restricts their applicability, necessitating denoising for optimal utilization. The existing deep learning (DL)-based denoising methods suffer from various limitations. For instance, convolutional neural networks (CNNs) struggle with long-range dependencies, while vision transformers (ViTs) struggle to capture local details. This article introduces a novel method, *UNFOLD*, that addresses these inherent limitations by harmoniously integrating the strengths of 3-D U-Net, 3-D CNN, and 3-D Transformer architectures. Unlike several existing methods that predominantly capture dependencies either along the spatial or the spectral dimension, *UNFOLD* addresses HSI denoising as a 3-D task, synergizing spatial and spectral information through the utilization of 3-D Transformer and 3-D CNN. It employs the self-attention (SA) mechanism of Transformers to capture the global dependencies and model long-range relationships across spatial and spectral dimensions. To overcome the limitations of 3-D Transformer in capturing fine-grained local and spatial features, *UNFOLD* complements it by incorporating 3-D CNN. Moreover, *UNFOLD* utilize a modified form of 3-D U-Net architecture for HSI denoising, wherein it employs a 3-D Transformer-based encoder instead of the conventional 3-D CNN-based encoder. It further capitalizes on the property of U-Net to integrate features across various scales, thereby enhancing efficacy by preserving intricate structural details. Results from extensive experiments demonstrate that *UNFOLD* outperforms the state-of-the-art HSI denoising methods.**

***Index Terms*— 3-D convolutional neural networks (CNNs), 3-D transformers, 3-D U-Net, hyperspectral imaging denoising, spatial spectral fusion.**

I. Introduction

HYPERSPECTRAL image (HSI) contains information at several spectrums, thus, extensively used in several real-world domains, including remote sensing [1], classification [2], [3], [4], agriculture [5], and marine monitoring [6]. It is represented as a 3-D array, incorporating two spatial and one spectral dimension. Unfortunately, noise can be added during the HSI sensing due to various factors, including limited light, photon effects, and atmospheric interference [7], thereby degrading HSI quality. This issue is mitigated by HSI denoising. In computer vision, image denoising is performed by analyzing each pixel's behavior with respect to its local neighborhood or global context. Several linear filters are extensively employed in the literature for local neighborhood analysis. Similarly, the nonlocal means filter [8] and non-local meets Global (NGMeet) [9] have been utilized to analyze global context or long-range dependencies for denoising [10]. Hence, image denoising can be effectively performed by analyzing the local neighborhood and global context.

Traditional HSI denoising was performed using denoising techniques for grayscale or RGB images. These methods applied denoising techniques on individual bands, like block-matching 3-D (BM3D) [11], focusing solely on spatial aspects and neglecting spectral data, causing spectral distortions. This limitation was addressed by introducing spatial-spectral-based methods. For instance, Othman and Qian [12] proposed hybrid spatial-spectral noise reduction (HSSNR), incorporating wavelet shrinkage to leverage spatial and spectral information. HSI can be seen as multiple 2-D images, leading to tensor-based denoising methods treating HSI as a 3-D tensor, such as TV-regularized low-rank tensor decomposition (LRTDTV) introduced by Wang et al. [13]. While these methods perform well in simpler scenarios, they struggle with complex mixed noise, highlighting the need for a robust and efficient solution to address these challenges [14].

Currently, the deep learning (DL) methods outperform the traditional hand-crafted features-based methods in solving several real-world problems, and HSI denoising also demonstrates similar behavior [7], [15], [16]. In this direction, convolutional neural networks (CNNs) have been extensively studied to exploit both spatial and spectral similarities for HSI denoising [7]. Moreover, these networks are augmented with U-Net [17], which is a de-facto for image translation tasks, like image denoising and segmentation. The U-Net networks are used due to their ability to effectively preserve the structural details using skip connections [15]. Nevertheless, the CNN-based HSI denoising methods exhibit limited performance because they fail to capture long-range dependencies in an image [18].

Apart from CNNs, the applicability of vision transformers (ViTs) has also been explored in HSI denoising [19]. The ViTs have outperformed CNNs in several computer vision tasks [20] as they effectively capture long-range dependencies using the self-attention (SA) mechanism. However, ViTs have limited performance, as they treat inputs as 1-D sequences. Consequently, while they excel at learning global contextual information, they may overlook local details. Thus, when the

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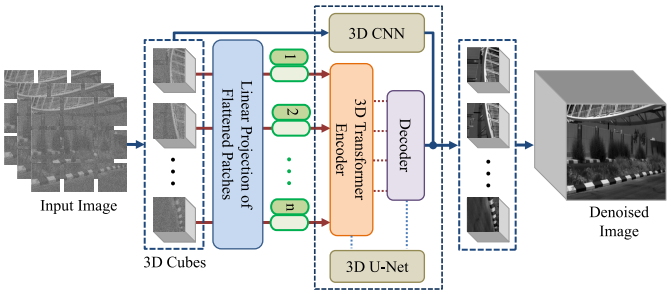


Fig. 1. Overview of *UNFOLD*. It utilizes a 3-D Transformer for capturing long-range dependencies, a 3-D U-Net for preserving structural integrity, and 3-D CNN for preserving local details for effective HSIs denoising.

global information acquired through Transformers is integrated with the local details captured by CNNs, it can augment the acquired features [21].

In this article, we leverage the strengths of 3-D U-Net, 3-D CNN, and 3-D Transformer architectures to mitigate the aforementioned issues by proposing a novel HSI denoising method called 3-D **U**-Net, 3-D C**N**N, and 3-D Trans**f**o**r**mer-based hyperspectra**l** image **d**enoising (*UNFOLD*). It reformulates HSI denoising as a 3-D denoising task and employs a 3-D transformer at the encoder to extract global contextual information. Since the Transformer-based encoder fails to capture local details efficiently, it is combined with 3-D CNN in parallel to consider local details effectively (see Fig. 1). Moreover, we combine the encoder and decoder using 3-D U-Net to preserve the structural details. In summary, our main contributions are given as follows.

1) To the best of our knowledge, we are the first one to introduce a novel HSI denoising method that comprises of 3-D U-Net, 3-D CNN, and 3-D Transformer.
2) Our proposed method systematically combines 3-D U-Net, 3-D CNN, and 3-D Transformer to leverage the strengths of each individual method. Essentially, it utilizes a 3-D transformer-based encoder for capturing long-range and depthwise dependencies, a 3-D U-Net for preserving structural integrity, and a 3-D CNN-based decoder to consider local details for effective HSI denoising.
3) Our proposed novel method is capable of handling noisy images with varying levels of noise intensity and excels at real-world scene noise removal without requiring specific parameter tuning. Furthermore, the application of our proposed model results in a marked enhancement in the performance of HSI classification.

The remainder of this article is organized as follows. Section II presents an overview of the existing research on HSI denoising. Section III provides the framework and specific details of our proposed method. Section IV analyzes and discusses the results obtained from our experiments. Section V concludes this work by summarizing the main findings and contributions.

II. Related Work

A. Traditional Methods for HSI Denoising

Traditional image denoising methods [22], [23], [24] perform noise removal using iterative optimization and hand-crafted priors. For instance, Chang et al. [25] uses low-rank method to leverage the inherent spatial and spectral similarity of HSIs. Likewise, NGMeet [9], combines the spatial and spectral similarities for HSI denoising. These methods are incompetent when the input image contains mixed noise, so they offer limited performance.

B. CNN-Based Methods for HSI Denoising

Usually, CNN outperforms traditional methods for computer vision tasks [26] including HSI denoising. For example, trainable spectral-spatial sparse coding (T3SC) [27] employs CNNs for spatial-spectral sparse coding and outperforms the traditional methods. Initially, CNN-based methods for HSI denoising were influenced by 2-D image denoising techniques. For instance, global reasoning network (GRNet) [15], combines several 2-D CNN networks sequentially for analyzing spatial-spectral similarities along with local and global information for HSI denoising. However, 3-D convolution is well-suited for handling HSIs as it simultaneously analyzes spatial and spectral dimensions. It is leveraged in hyperspectral image deep convolutional neural network (HSID-CNN) [7], where 2-D CNN and 3-D CNN networks are used for spatial and spectral analysis, respectively. However, better performance is provided by 3-D quasi-recurrent neural network (QRNN3D) [16] which analyzes the spatial-spectral correlation using 3-D CNN.

C. Transformer Methods for HSI Denoising

Recently, Transformer architecture [20] has been introduced, which incorporates a global SA mechanism. This mechanism allows efficient capture of long-range dependencies in the sequential data. In computer vision, ViT [19], is extensively studied [28]. It divides an image into nonoverlapping patches and effectively captures the global relationships among these local patches to address the limitations of CNNs. However, if 2-D transformer is used for HSI denoising, it alters the original 3-D structure of HSIs, like global correlation along the spectrum (GCS) or spatial self-similarity. This issue can be addressed using a 3-D Transformer for HSI denoising that captures spatial-spectral global dependence. In this direction, Wang et al. [29] introduces the [spatial-spectral transformer denoising (SSTD)] network, which explores the local and global dependencies in HSIs. Likewise, spatial-spectral transformer (SST) [30] employs attention mechanisms to analyze the spatial and spectral similarities for HSI denoising. Similarly, Li et al. [31] introduced the SA module for spatial similarity and the spectral enhancement module to extract informative vectors from HSI patches and suppress the noise. Unfortunately, transformer networks suffer from an inherent problem of overlooking the local information [14]. We mitigate it by combining our 3-D transformer with our 3-D CNN and 3-D U-Net.

III. Proposed Method

In this section, we present our novel method for HSI denoising, which capitalizes on the strengths of three key architectural components: the 3-D Transformer, the 3-D CNN, and the 3-D U-Net (refer Fig. 2). The foundation of our approach is the 3-D Transformer, which excels at capturing global dependencies in (3-D) data. Its SA mechanism

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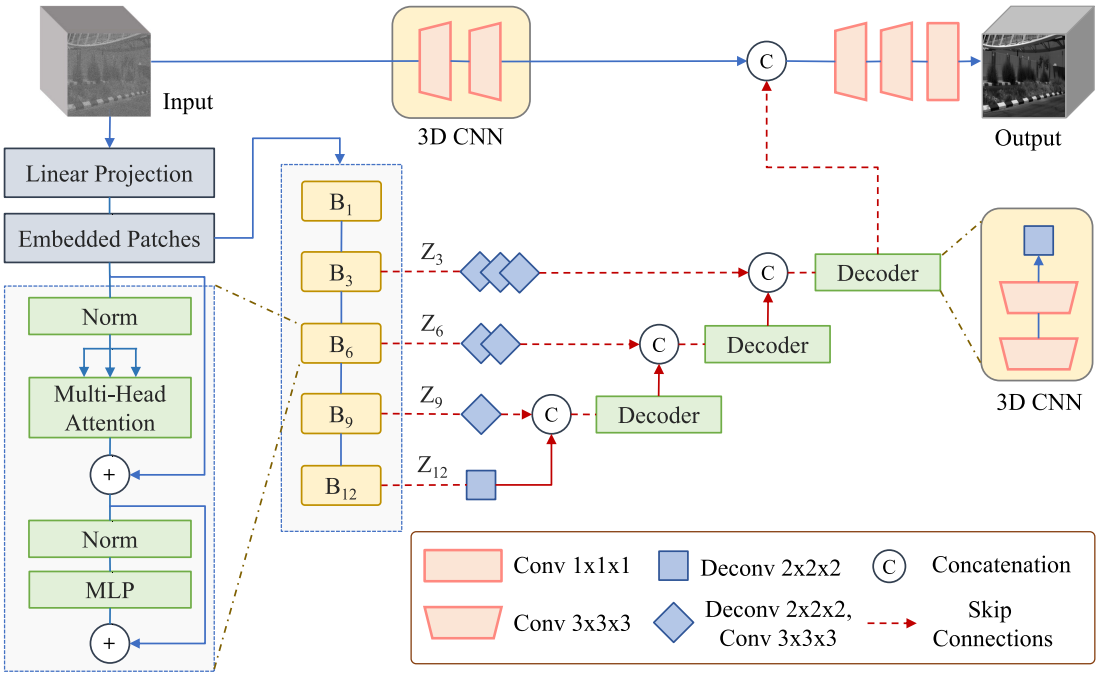


Fig. 2. Detailed architecture of the proposed method *UNFOLD*. The input is divided into cubes and passed through a 3-D Transformer that acts as an encoder of the 3-D U-Net. Encoder outputs from different Transformer blocks are combined with the decoder through skip connections. To preserve local spatial information, the input is also passed through a 3-D CNN, the output of which, is merged with the decoder output to obtain the denoised HSI.

allows it to model long-range relationships across spatial and spectral dimensions, facilitating the extraction of meaningful features. However, Transformers exhibit limitations in precisely capturing fine-grained local and spatial features. Hence, we complement the Transformer by incorporating 3-D CNN to adeptly capture local patterns and spectral intricacies. We also employ a modified form of 3-D U-Net architecture that comprises an encoder pathway that extracts features by downsampling the input, followed by a decoder pathway that generates an output by upsampling the features. 3-D U-Net also employs skip connections that connect corresponding layers between the encoder and decoder pathways, facilitating the integration of features from different scales and hence preserving the structural details.

A. 3-D Transformer

The input for the standard ViT [19] is a 2-D image $\mathbf{X} \in \mathbb{R}^{h\times w\times c}$, where h is the height, w is the width, and c is the number of channels of the image. The image $\mathbf{X}$ is then divided into fixed sized nonoverlapping flattened patches $\mathbf{x}_p \in \mathbb{R}^{m\times(p^2\cdot c)}$. Kindly note that each patch has a size of $p\times p$, which, when flattened, becomes a 1-D sequence of size p^2. The total number of patches, m, is given by $m = {}^{h\cdot w}/_{p^2}$.

However, in the context of the HSI denoising task, the input is a 3-D volume, $\mathbf{X} \in \mathbb{R}^{h\times w\times c\times d}$, that has an additional dimension, d, which is the number of spectral bands present in the HSI. Similar to the standard ViT, the proposed method divides the input volume $\mathbf{X}$ into smaller fixed sized nonoverlapping 3-D cubes, which are then flattened into 1-D vectors, $\mathbf{x}_v \in \mathbb{R}^{n\times(p^3\cdot c)}$. Here, the size of each cube is $p\times p\times p$, and the total number of cubes, is given by $n = {}^{(h\cdot w\cdot d)}/_{p^3}$. These flattened cubes are then mapped to a đ-dimensional space using a trainable linear projection, $\mathbf{E} \in \mathbb{R}^{(p^3\cdot c)\times đ}$, to produce cube embeddings. To inject and preserve positional information, the cube embeddings are augmented with standard learnable 1-D positional embeddings $\mathbf{E}_{\text{pos}} \in \mathbb{R}^{n\times đ}$, to position aware cube embeddings $\mathbf{e}_0$. Formally, it can be expressed as

$$\mathbf{e}_0 = \left[\mathbf{x}_v^1\mathbf{E}; \mathbf{x}_v^2\mathbf{E}, \ldots, \mathbf{x}_v^n\mathbf{E}\right] + \mathbf{E}_{\text{pos}}. \tag{1}$$

After the embedding layer, we employ a series of 3-D Transformer blocks, which consist of a multihead SA (MSA) operation followed by a multilayer perceptron (MLP). Mathematically, it can be defined as

$$\tilde{\mathbf{e}}_\ell = \text{MSA}(\text{LayerNorm}(\mathbf{e}_{\ell-1})) + \mathbf{e}_{\ell-1}, \quad \ell = 1, \ldots, L \tag{2}$$

$$\mathbf{e}_\ell = \text{MLP}\big(\text{LayerNorm}\big(\tilde{\mathbf{e}}_\ell\big)\big) + \tilde{\mathbf{e}}_\ell, \quad \ell = 1, \ldots, L \tag{3}$$

where, the operator LayerNorm$(\cdot)$ denotes the layer normalization opertion, ℓ denotes the ℓth block, out of the total L 3-D Transformer blocks. The MSA consists of m SA heads running in parallel. Each SA block constitutes a functional unit with learnable parameters that encapsulates the relation between a query vector $(\mathbf{q}_\ell)$ and its corresponding key $(\mathbf{k}_\ell)$ and value $(\mathbf{v}_\ell)$ vector projections of a given cube embedding $\mathbf{e}_\ell$. The attention weights $(\mathbf{A}_\ell)$ are calculated as

$$\mathbf{A}_\ell = \text{Softmax}\left(\frac{\mathbf{q}_\ell\mathbf{k}_\ell^\top}{\sqrt{đ_h}}\right) \tag{4}$$

where $đ_h = đ/m$ is scaling factor employed to stabilize gradients during training and to ensure balanced contributions from different attention heads. The SA is then computed as $\text{SA}_\ell = \mathbf{A}_\ell\mathbf{v}_\ell$, after which the MSA is computed by concatenating the SAs from m heads, followed by linearly transforming the concatenated vector using a learnable projection matrix

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$\mathbf{W}_{\text{msa}} \in \mathbb{R}^{m \cdot \text{đ}_h \times \text{đ}}$. Formally, the MSA at the ℓth block is calculated as

$$\text{MSA}_\ell = \left[\text{SA}_\ell^1; \text{SA}_\ell^2, \ldots, \text{SA}_\ell^m\right] \mathbf{W}_{\text{msa}} \tag{5}$$

where SA_ℓ^m is SA of the mth head at the ℓth block.

B. 3-D CNN Architecture

A 3-D CNN operates on 3-D data, enabling it to capture spatial and temporal correlations within a volumetric region. This is particularly advantageous when dealing with 3-D data such as HSIs, where the third dimension corresponds to different spectral bands. This is in contrast to the 2-D CNNs that process the data, without directly incorporating spectral information. A 2-D CNN processes each band d independently across 2-D spatial patches, whereas a 3-D CNN processes multispectral information over a 3-D cube, thus inherently capturing intricate spatial-spectral patterns in HSIs. This distinction allows 3-D CNNs to effectively exploit spatial and spectral features in tasks like HSI denoising. For a given input $\mathbf{X}$, the 2-D convolution operation for dth band is performed as

$$\mathbf{Y}_d[i, j] = \mathbf{b}[i, j] + \sum_{m=-\alpha}^{\alpha} \sum_{n=-\beta}^{\beta} \mathbf{X}[i+m, j+n] \cdot \mathbf{W}[m, n] \tag{6}$$

where Y is the output feature map; W denotes the learnable 2-D convolutional kernel; and α, β determine the receptive fields in the width and height dimensions, respectively. Similarly, the 3-D convolution operation for the input X can be represented as follows:

$$\mathbf{Y}[i, j, k] = \mathbf{b}[i, j, k] + \sum_{n=-\beta}^{\beta} \sum_{p=-\gamma}^{\gamma} \times \sum_{m=-\alpha}^{\alpha} \mathbf{X}[i+m, j+n, k+p] \cdot \mathbf{W}[m, n, p]$$

where W denotes the learnable 3-D convolutional kernel; and α, β, γ determine receptive fields along the three dimensions.

C. 3-D U-Net Architecture

The 3-D U-Net [32] is an extension of the widely used 2-D U-Net architecture [17], specifically designed for processing 3-D data, such as 3-D medical image segmentation [33], [34]. Similar to its 2-D counterpart, the 3-D U-Net combines a U-shaped architecture with skip connections to efficiently capture both low-level and high-level features. In our work, we propose a modified 3-D U-Net architecture tailored for HSI denoising in which we employ a 3-D Transformer (see Section III-A)-based encoder instead of the traditional 3-D CNN-based encoder (see Section III-B).

In this modified architecture, we take the outputs from specific blocks of the 3-D Transformer, $\ell \in$ 3rd, 6th, 9th, and 12th, to obtain cube embedding vectors $\mathbf{e}_\ell \in \mathbb{R}^{(h \cdot w \cdot d)/p^3 \times \text{đ}}$ and reshape them into 3-D volumes $\hat{\mathbf{e}}_\ell \in \mathbb{R}^{(h/p) \times (w/p) \times (d/p) \times \text{đ}}$. These reshaped volumes $\hat{\mathbf{e}}_\ell$ are then processed through a series of $2 \times 2 \times 2$ deconvolution layers and $3 \times 3 \times 3$ convolution layers to obtain encoder outputs at desired resolutions. The encoder output at the last block (that is $\ell = 12$), is passed through a single $2 \times 2 \times 2$ deconvolution layer, the output of which is then concatenated with the encoder output of the previous block (that is $\ell = 9$). The concatenated output is then processed through a series of $3 \times 3 \times 3$ convolution and $2 \times 2 \times 2$ deconvolution layers to obtain a volume of resolution match to the encoder output previous block (that is $\ell = 6$). This procedure is reiterated for each preceding layer until we attain the original input resolution. At the end, the final output is passed through a $1 \times 1 \times 1$ convolutional layer, which uses a softmax activation function to obtain the denoised HSI.

D. Training Loss Function

We use a structural similarity index measure (SSIM) [36] loss as a loss function as used in [16], [30], and [31]. It measures image similarity by considering luminance, contrast, and structure. The formula for SSIM is defined as follows:

$$\text{SSIM}(X, Y) = \frac{(2 * \mu_X * \mu_Y + c_1) * (2 * \sigma_{XY} + c_2)}{\left(\mu_X^2 + \mu_Y^2 + c_1\right) * \left(\sigma_X^2 + \sigma_Y^2 + c_2\right)} \tag{7}$$

where X and Y are the two images being compared; μ_X and μ_Y are the average values of X and Y, respectively; σ_X^2 and σ_Y^2 are the variances of X and Y, respectively; σ_{XY} is the covariance of X and Y; c_1 and c_2 are constant values that prevent instability when the denominator approaches zero.

IV. Experimental Results

A. Experimental Settings

1) Dataset: We used the ICVL hyperspectral dataset [37], which consists of 201 images with a spatial resolution of 1392×1300 and 31 spectral bands spanning from 400 to 700 nm. Out of this, 100 images were allocated for training, 20 for validation, and the rest for testing. The images are normalized to [0, 1], before passing it to the model. For experimentation, we have used the same methodology prevalent in HSI denoising and proposed in [38]. That is, we cropped the images into multiple overlapping cubes with dimensions $64 \times 64 \times 31$ and $512 \times 512 \times 31$ for training and testing, respectively. Likewise, we evaluated the efficacy of *UNFOLD* by performing HSI denoising experiments on real-world data samples taken from the University of Pavia (PU) [39], Indian Pines (IP) [40], and Urban [41] datasets.

2) Noise Settings: HSIs captured in the real world are often affected by Gaussian noise, stripe noise, dead pixels, and impulse noise [42], [43]. For thorough experimentations, we simulate the noisy HSIs using the following noise patterns [16]: 1) i.i.d Gaussian noise ranging from level 10 to level 70, with stepsizes of 20 and 2) five complex noise scenarios, namely, Non-i.i.d Gaussian noise, Gaussian + stripe noise, Gaussian + deadline noise, Gaussian + impulse noise, and mixture noise.

3) Implementation Details: We performed our experiments with PyTorch[1] and MONAI[2] on a PC with NVIDIA GeForce

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[2] https://monai.io/

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TABLE I
PERFORMANCE OF *UNFOLD* UNDER GAUSSIAN NOISE ON THE ICVL DATASET

	$\sigma = 10$			$\sigma = 30$			$\sigma = 50$			$\sigma = 70$		
Methods ↓	PSNR	SSIM	SAM	PSNR	SSIM	SAM	PSNR	SSIM	SAM	PSNR	SSIM	SAM
Noisy	28.13	0.8792	18.72	18.59	0.5523	37.90	14.15	0.3476	49.01	11.23	0.2301	56.45
BM4D [35]	40.78	0.9930	2.99	37.69	0.9872	5.02	34.96	0.9850	6.81	33.15	0.9554	8.40
LLRT [25]	46.72	0.9983	1.60	41.12	0.9920	2.52	38.24	0.9830	3.47	36.23	0.9732	4.46
NGMeet [9]	47.90	0.9988	1.39	42.44	0.9816	2.06	39.69	0.9658	2.49	38.05	0.9531	2.83
HSID-CNN [7]	43.14	0.9918	2.12	40.30	0.9854	3.14	37.72	0.9746	4.27	34.95	0.9521	5.84
GRNet [15]	45.25	0.9976	1.83	42.09	0.9957	2.18	40.25	0.9936	2.42	38.95	0.9914	2.63
QRNN3D [16]	45.61	0.9977	1.80	42.18	0.9955	2.21	40.05	0.9929	2.63	38.09	0.9883	3.42
T3SC [27]	45.81	0.9979	2.02	42.44	0.9957	2.44	40.39	0.9933	2.85	38.80	0.9904	3.26
SST [30]	48.28	0.9990	1.30	43.32	0.9971	1.87	41.09	0.9950	2.19	39.55	0.9921	2.46
SERT [31]	47.72	0.9988	1.36	43.56	0.9969	1.77	41.33	0.9949	2.05	39.82	0.9929	2.30
OURS	**48.52**	**0.9991**	**1.28**	**44.81**	**0.9973**	**1.74**	**44.37**	**0.9968**	**1.99**	**43.43**	**0.9956**	**2.12**

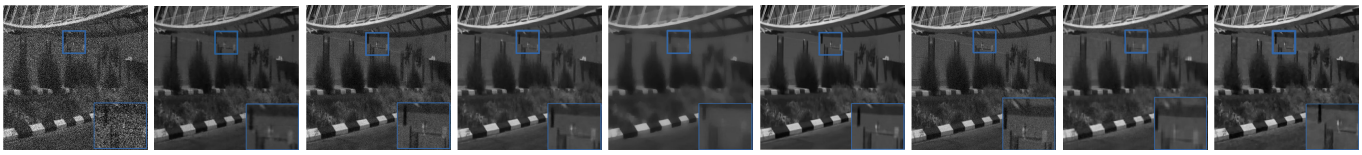


Fig. 3. Visual comparisons on a denoised image from the ICVL dataset under the Gaussian noise with a noise level $\sigma = 50$. (a) Noisy. (b) BM4D. (c) LLRT. (d) NGMeet. (e) HSID. (f) GRNet. (g) QRNN3D. (h) T3SC. (i) Ours.

RTX 3050 GPU, AMD Ryzen 5 5600X Six-Core Processor, and 16-GB RAM. The AdamW optimizer [44] was utilized with an initial learning rate of 0.0001, employing a batch size of 4 for 500 epochs. The total training duration was approximately 15 h. The standard ViT architecture (ViT-B16 [19]) was employed without the implementation of any data augmentation techniques.

4) Competing Methods: We assessed the performance of our method in comparison to traditional and DL methods under various noise conditions, including Gaussian and complex noise. Generally, traditional methods perform better when the noise scenarios are known. In contrast, DL methods can tackle a variety of noises by training multiple models for different noises. Essentially, we compare *UNFOLD* with the following state-of-the-art methods, which are filtering-based BM4D [35], tensor-based hyper-laplacian regularized unidirectional low-rank tensor (LLRT) [25], and orthogonal basis-based NGMeet [9]. DL-based methods, (HSID-CNN [7], GRNet [15], QRNN3D [16], T3SC [27], SST [30], and spectral enhanced rectangle transformer (SERT) [31]).

5) Quantitative Metrics: Just like the other existing works in HSI denoising [16], our performance is evaluated using peak signal-to-noise ratio (PSNR) and structural similarity index [45] for measuring spatial-based similarities, and spectral angle mapper (SAM) [46] for measuring spectral similarity.

B. Simulated Experimental Results

1) Gaussian Noise: In this scenario, the HSI is corrupted by adding zero-mean additive white Gaussian noise at different levels of variance (σ). The quantitative outcomes are presented in Table I. It indicates that *UNFOLD* outperforms other well-known methods, with improvements in PSNR of more than 1.2, 2.8, and 3.1 dB for noise levels of 30, 50, and 70, respectively. Furthermore, it depicts that DL and traditional methods demonstrate similar performance for high noise levels but DL methods exhibit better performance than traditional methods for low noise levels. In contrast, our proposed method consistently provides superior results, even for low noise levels showcasing its robustness and generalization capabilities. For better visualization, refer to Fig. 3 that depicts the denoised HSIs obtained using considered methods, corresponding to the noise level of 50. It reveals that our proposed method successfully mitigates Gaussian noise while accurately preserving the underlying HSI structure. To visualize the efficacy of different denoising methods in the spectral domain, Fig. 4 provides the spectral profiles. It depicts that *UNFOLD* surpasses all other methods in terms of spectral performance and closely approximates the ground truth. Furthermore, the corresponding training convergence curve for *UNFOLD* is shown in Fig. 5, illustrating the model's training stability.

2) Complex Noises: Table II presents the quantitative results of our proposed method and other compared methods on the ICVL dataset, and the visual comparisons under different noise conditions are shown in Figs. 6–8. Notably, it shows that *UNFOLD* outperforms other methods in handling various noise degradations. However, traditional methods are incompetent for impulse and mixed noise. It happens because such noises result in the loss of relevant spatial and spectral information that handcrafted priors cannot interpret without knowing the noise characteristics [30]. Nevertheless, DL methods achieve more promising results for such noise due to

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TABLE II
PERFORMANCE OF *UNFOLD* UNDER FIVE DIFFERENT COMPLEX NOISES ON THE ICVL DATASET

Noise Type →	Non-i.i.d Gaussian			Gaussian+Stripe			Gaussian+Deadline			Gaussian+Impulse			Gaussian+Mixture		
Methods ↓	PSNR	SSIM	SAM	PSNR	SSIM	SAM	PSNR	SSIM	SAM	PSNR	SSIM	SAM	PSNR	SSIM	SAM
Noisy	18.29	0.5116	46.20	17.51	0.4867	46.98	17.50	0.4770	47.55	14.93	0.3758	46.98	13.91	0.3396	51.53
BM4D [35]	36.18	0.9767	5.78	35.63	0.9730	6.26	33.77	0.9615	6.85	29.79	0.8613	21.59	28.01	0.8419	23.59
LLRT [25]	34.18	0.9618	4.88	34.27	0.9628	4.93	32.98	0.9559	5.29	28.85	0.8819	18.17	28.06	0.8697	19.37
NGMeet [9]	34.90	0.9745	5.37	34.88	0.9660	5.42	33.41	0.9665	6.55	27.02	0.7884	31.20	26.13	0.7796	31.89
HSID-CNN [7]	39.28	0.9819	3.80	38.09	0.9765	4.59	38.33	0.9783	3.99	36.21	0.9663	5.48	35.30	0.9588	6.29
GRNet [15]	35.19	0.9780	5.19	34.85	0.9772	5.41	33.78	0.9744	5.42	32.78	0.9606	8.26	30.91	0.9617	8.26
QRNN3D [16]	42.18	0.9950	2.84	41.68	0.9943	2.97	41.69	0.9942	2.61	40.32	0.9914	4.31	39.08	0.9892	4.80
T3SC [27]	41.95	0.9922	4.18	41.32	0.9937	3.27	39.59	0.9924	4.86	37.85	0.9843	6.53	35.53	0.9767	8.12
SST [30]	43.57	0.9970	2.05	42.97	0.9960	2.15	42.74	0.9961	2.21	41.66	0.9950	2.60	38.78	0.9911	2.99
SERT [31]	44.20	0.9971	1.69	43.68	0.9969	1.97	43.66	0.9969	1.99	42.67	0.9959	2.30	40.00	0.9937	2.84
OURS	**46.13**	**0.9973**	**1.63**	**44.70**	**0.9974**	**1.96**	**45.67**	**0.9975**	**1.99**	**44.24**	**0.9965**	**2.27**	**44.50**	**0.9944**	**2.79**

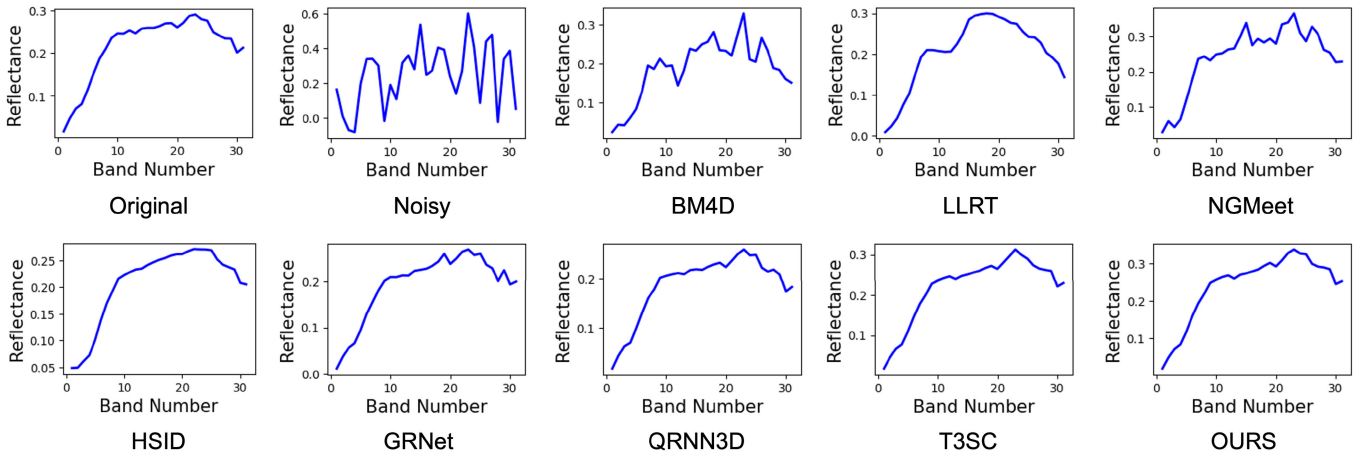


Fig. 4. Reconstructed spectral curves using various methods at the spatial coordinates (150, 150), under Gaussian noise ($\sigma = 50$) on the ICVL dataset. The vertical axis, labeled "Reflectance," represents the bandwise values of pixels at the same location, while the horizontal axis corresponds to the band number.

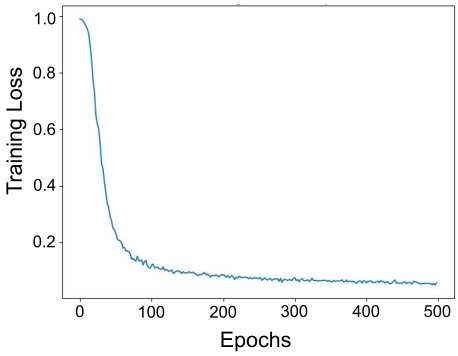


Fig. 5. Training convergence curves of *UNFOLD*.

appropriate training. Hence, we also leverage the DL methods to mitigate such noises.

C. Real-Data Experimental Results

To reaffirm the efficacy of our proposed method, we also conduct real-data experiments using three real-world HSI datasets: IP, Urban, and PU. The IP dataset comprises 145 × 145 pixels along with 220 spectral bands; the Urban dataset comprises 307 × 307 pixels with 210 spectral bands; and PU dataset comprises a total of 103 spectral bands with an image resolution of 610 × 340 pixels. The qualitative denoising results are shown in Fig. 9 for different band combinations. It can be observed from Fig. 9 that *UNFOLD* achieves better performance in terms of visual quality than the existing methods.

For a more rigorous evaluation, we compared the classification accuracy of the HSI both before and after denoising using various methods. For consistency, support vector machine (SVM) classifier [1] is employed for classification and the results are assessed using the overall accuracy (OA) and the kappa coefficient as evaluation metrics. Fig. 10 illustrates the SVM classification outcomes for the IP, Urban, and PU datasets, while the detailed quantitative classification results are presented in Table III. It can be observed from Table III that all the compared methods and *UNFOLD* improve classification accuracy postdenoising. Among these methods, *UNFOLD* emerges as the top-performing method, achieving the best OA and kappa values.

D. Comparison With Transformer Models for RGB Denoising

For comprehensiveness, we have thoroughly compared our proposed method with state-of-the-art transformer-based methods for denoising RGB images, as presented in Table IV. Table IV highlights that *UNFOLD* achieved the best results in HSI denoising. This behavior can be attributed to existing methods focusing on capturing dependencies along either the spatial dimension [47], [48] or the spectral dimension [49].

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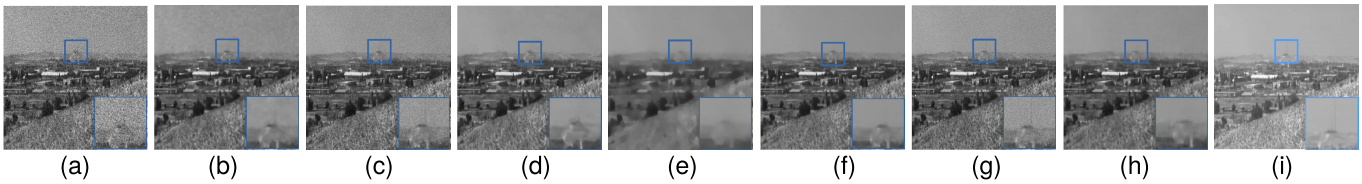


Fig. 6. Visual comparisons on a denoised image from the ICVL dataset under the complex noise (Case 1). (a) Noisy. (b) BM4D. (c) LLRT. (d) NGMeet. (e) HSID. (f) GRNet. (g) QRNN3D. (h) T3SC. (i) Ours.

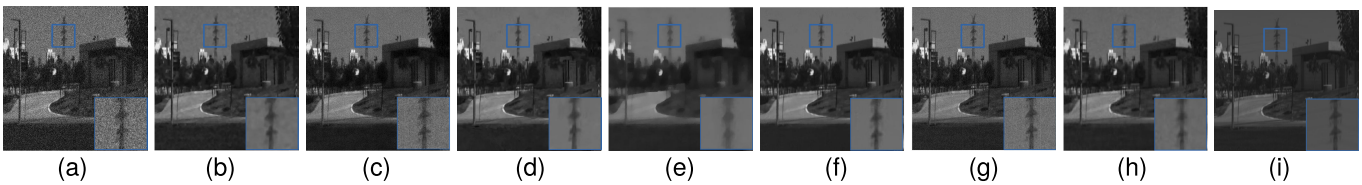


Fig. 7. Visual comparisons on a denoised image from the ICVL dataset under the complex noise (Case 2). (a) Noisy. (b) BM4D. (c) LLRT. (d) NGMeet. (e) HSID. (f) GRNet. (g) QRNN3D. (h) T3SC. (i) Ours.

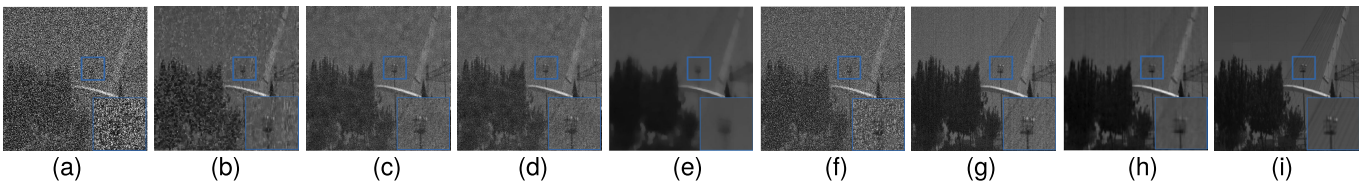


Fig. 8. Visual comparisons on a denoised image from the ICVL dataset under the complex noise (Case 5). (a) Noisy. (b) BM4D. (c) LLRT. (d) NGMeet. (e) HSID. (f) GRNet. (g) QRNN3D. (h) T3SC. (i) Ours.

TABLE III

SVM CLASSIFICATION RESULTS ON THE IP, URBAN, AND PU DATASETS BEFORE AND AFTER HSI DENOISING

Dataset	**Index**	Noisy	[35]	[25]	[9]	[7]	[15]	[16]	[27]	[30]	[31]	**OURS**
IP	OA	71.25	82.80	81.99	83.01	83.31	78.66	84.38	85.62	87.01	88.17	**89.92**
	Kappa	0.6696	0.8040	0.7952	0.8130	0.8101	0.7560	0.8228	0.8366	0.8422	0.8628	**0.8837**
Urban	OA	86.11	91.05	91.27	92.33	92.76	90.01	92.99	93.47	94.17	94.91	**95.13**
	Kappa	0.8385	0.8890	0.8847	0.8909	0.9088	0.8863	0.9086	0.9115	0.9189	0.9267	**0.9405**
PU	OA	86.54	91.72	91.88	92.25	92.81	90.45	93.10	94.84	95.67	96.52	**97.82**
	Kappa	0.8068	0.9012	0.9007	0.9113	0.9167	0.8891	0.9218	0.9330	0.9376	0.9588	**0.9698**

TABLE IV

COMPARISON WITH TRANSFORMERS-BASED MODELS DESIGNED FOR DENOISING OF RGB IMAGES, UNDER GAUSSIAN NOISE ($\sigma = 50$) ON ICVL DATASET

Metric	SwinIR [47]	Restormer [49]	CSwin [48]	**Ours**
PSNR	40.44	41.07	42.04	**44.37**
SSIM	0.9938	0.9945	0.9951	**0.9968**
SAM	2.32	2.05	2.18	**1.99**

In contrast, *UNFOLD* jointly captures spatial and spectral dependencies, leveraging local and global information using a 3-D transformer and a 3-D CNN. Additionally, it employs a 3-D U-Net that integrates features across various scales, thereby effectively preserving intricate structural details.

V. ABLATION STUDY

We conducted thorough ablation studies using HSI Gaussian denoising to assess the efficacy of individual subcomponents within our *UNFOLD* method. The subcomponents are: 3-D U-Net used to preserve the structural similarities; 3-D CNN used to consider the local details; and 3-D transformer used to analyze long-range and depthwise dependencies. We perform the experiments by removing each subcomponent from our proposed method, and the corresponding evaluation metrics, PSNR and SSIM, are provided in Table V.

TABLE V

BEHAVIOR OF DIFFERENT COMPONENTS OF *UNFOLD*, UNDER GAUSSIAN NOISE ON THE ICVL DATASET

Components ↓			$\sigma = 30$		$\sigma = 50$		$\sigma = 70$	
CNN	Trans-former	U-NET	PSNR	SSIM	PSNR	SSIM	PSNR	SSIM
2D	✗	✗	36.89	0.9688	35.42	0.9622	34.05	0.9522
3D	✗	✗	38.28	0.9731	37.23	0.9617	36.33	0.9551
✗	3D	3D	43.35	0.9810	42.54	0.9743	41.50	0.9713
3D	3D	✗	42.86	0.9721	42.14	0.9712	40.88	0.9637
3D	✗	3D	42.32	0.9709	40.25	0.9615	38.60	0.9602
3D	2D	3D	41.79	0.9893	39.83	0.9847	37.96	0.9849
3D	3D	3D	**44.81**	**0.9973**	**44.37**	**0.9968**	**43.43**	**0.9956**

Table V demonstrates a significant performance degradation when 2-D CNN is used instead of 3-D CNN. It can be attributed to the neglection of the spectral correlations in 2-D CNN. The same behavior is observed in [50], which advises utilizing the 3-D CNN instead of the 2-D CNN because

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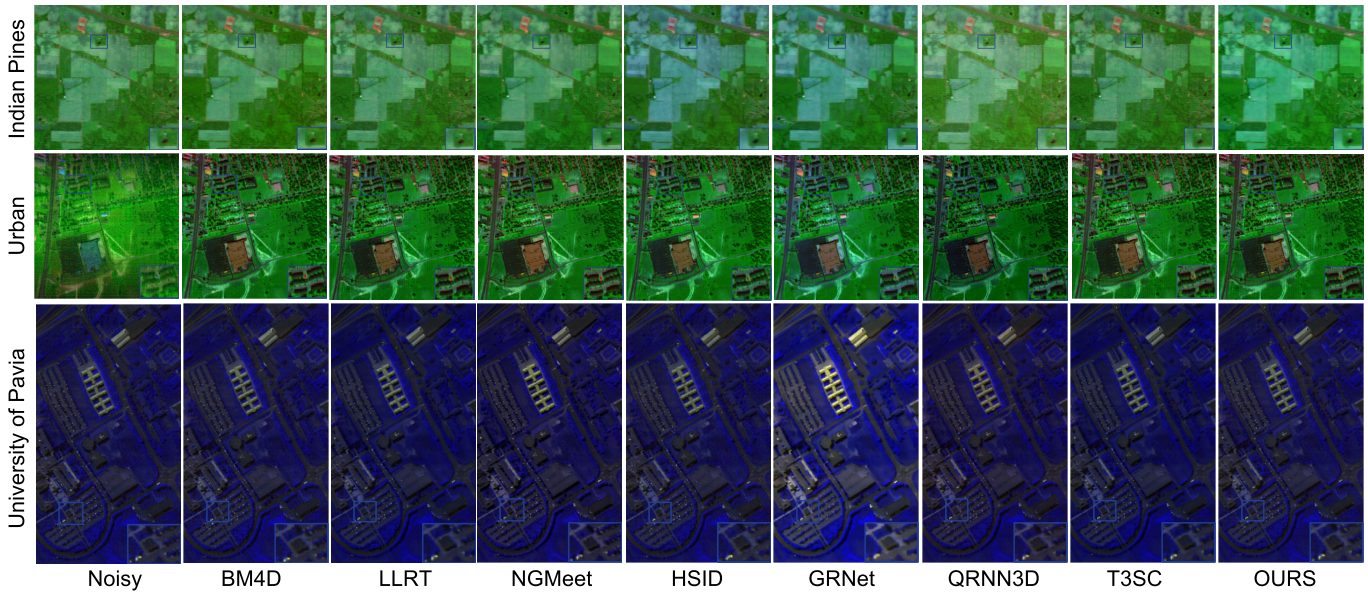


Fig. 9. In real-world experiments, IP, Urban, and PU datasets were used to obtain denoising results for bands (3, 102, 210), (1, 103, 204), and (2, 4, 95), respectively.

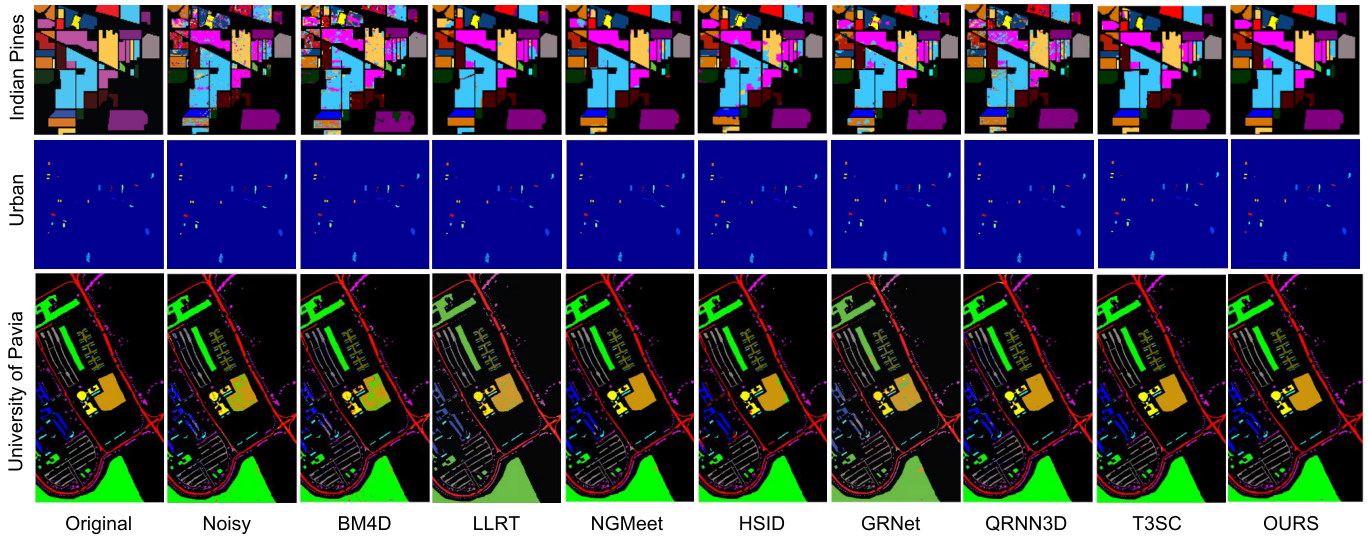


Fig. 10. SVM classification results for IP, Urban, and PU datasets.

2-D CNN cannot completely process the 3-D spectral-spatial structure in the HSI. We observe similar characteristics with transformers; that is, the performance degrades when a 2-D transformer replaces the 3-D transformer because the 2-D transformer fails to capture the spectral correlation and depth-wise information, as stated in [29]. Our method uses 3-D CNN and 3-D transformer in parallel to analyze local and global details, respectively. In our ablation study, removing the 3-D CNN subcomponent refers to removing the 3-D CNN from the parallel branch while keeping the 3-D CNN in the decoder intact. It can be inferred from Table V that 3-D CNN provides inferior efficacy than 3-D transformer because it neglects long-range dependence. Nevertheless, it provides local information, which is neglected when only a 3-D transformer is used. Hence, we observe performance improvement in Table V when both 3-D CNN and 3-D transformer are used. Moreover, we observe the performance improvements when 3-D U-Net is utilized in the network because it effectively preserves the structural information between the encoder and decoder. As an epitome, it can be inferred that optimal performance can be achieved by incorporating all the analyzed subcomponents, and performance can be degraded if any subcomponent is avoided.

It is worth highlighting that our proposed approach, *UNFOLD*, demonstrates comparable computational efficiency when compared with state-of-the-art methods like SST [30] and SERT [31]. To provide specifics, *UNFOLD* takes an average inference time of 5.8 s, which is notably close to the method SST [30] taking around 7.4 s and method SERT [31] requiring roughly 3.2 s.

VI. Conclusion

This article introduced a novel method *UNFOLD*, for denoising HSI. It has effectively addressed the limitations

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inherent in existing DL-based denoising methods for HSIs by seamlessly integrating the strengths of 3-D CNN, 3-D Transformer, and 3-D U-Net architectures. Unlike conventional methods that often focus solely on spatial or spectral dependencies, *UNFOLD* treated it as a 3-D task to synergize both spatial and spectral information. *UNFOLD* successfully mitigated the limitations of 3-D Transformer in capturing fine-grained local and spatial features by incorporating 3-D CNN. Moreover, it utilized a modified form of 3-D U-Net architecture consisting of a 3-D Transformer-based encoder for HSI denoising. Furthermore, U-Net has improved our efficacy by integrating features across various scales. While our *UNFOLD* exhibits superior denoising performance compared to the state-of-the-art HSI denoising methods, it has a limitation that it utilizes 3-D convolutional blocks and 3-D transformer, which significantly increases model parameters as compared to SST [30] and SERT [31]. In the future, we will explore the development of a lightweight model to address this limitation. Also, we will tackle the data scarcity problem by investigating data augmentation methods and enhancing the adaptability and generalization of our *UNFOLD* method.

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